Bug Report: Deleting a Task of a Missing Project

Bug ID: PROJ-003

Summary: Deleting a task link from a nonexistent project returns a NullPointerException.

Description:

DELETE `/projects/:id/tasks/:id` with a project id that does not exist returns 400 with the text of an unhandled null pointer error whenever the todo is a task of some other project. The text depends on the Java version that runs the application: `java.lang.NullPointerException` on Java 11, and `Cannot invoke "...ThingInstance.getRelationships()" because "parent" is null` on Java 15 and later. The same happens on `/projects/:id/categories/:id`. When the todo has no project the API returns the correct 404.

Potential impact:

The response exposes an unhandled internal exception and reports a client error code for a resource that is simply missing. Clients cannot tell a missing project from a malformed request, and the unhandled path is a sign of missing validation in the delete handler.

Steps to reproduce:

1. Start the application with its default data, where todo 1 is a task of project 1.
2. Send DELETE `/projects/0/tasks/1`.

Expected result: 404 with the message "Could not find any instances with projects/0/tasks/1".

Actual result: 400 with the null pointer message described above. No data changes.

Found in: Projects session; curl commands in `Exploratory testing scripts/Projects/projects_session.sh`, section "Behavior that differs from the documentation".

Unit tests: `deleteTaskOfMissingProjectReturnsNotFound` (expected), `deleteTaskOfMissingProjectReportsANullPointerException` (actual). Expected tests are in `ProjectsDocumentedBehaviorTest`; actual tests are in `ProjectsActualBehaviorTest`.